George Thanasoulis

ELECTRICAL & COMPUTER ENGINEER

Thessaloniki, Greece | +30 698 5702 189 | geor.thanasoulis@gmail.com

LinkedIn Profile: Georgios Thanasoulis

GitHub Profile: https://github.com/GeorgeThan414

Objective

Electrical and Computer Engineer with a strong interest in Artificial Intelligence and its real-world applications. Currently involved in two European research projects, contributing to innovative solutions in data analysis and intelligent systems. Interested in part-type or contract opportunities where advanced machine learning and electrical engineering can be applied to real-world challenges.

Experience

Centre for Research & Technology Hellas (CERTH) | Data Scientist **02/2025 - Present**

Currently, participating in two European research projects focused on the design of AI algorithms for forecasting and optimizing resource orchestration in cloud and edge computing environments, and on generative models for synthetic data and image generation.

Responsibilities include:

Project 1: AI Forecaster – Cloud & Edge Telemetry Forecasting

- Designing an end-to-end forecasting pipeline for multivariate telemetry data (CPU, RAM, latency, energy).
- Developed multivariate time-series forecasting models in PyTorch, combining hybrid LSTM–Transformer architectures, FiLM-based conditioning mechanisms, and fine-tuned pre-trained transformer models (e.g., Chronos) for telemetry forecasting.
- Exposed inference via FastAPI REST APIs and containerized services using Docker.
- Deployed and orchestrated workloads on Kubernetes, with scheduled inference using Jobs and CronJobs and deployments.
- Integrated MinIO (S3-compatible storage) for telemetry data, models, and predictions.

Project 2: Generative AI for Synthetic Data & Image Generation

- Designed convolutional architectures for image reconstruction and generation.
- Implemented VAEs, Conditional VAEs, and GANs for synthetic data generation and augmentation.
- Developed time-series VAEs with Fourier-based priors for seasonally conditioned sequence generation.
- Utilized pre-trained diffusion models for image synthesis and latent-space analysis.

Education

Aristotle University of Thessaloniki| MSc in Artificial Intelligence **09/2025 – Present**

- Electronics and Information Systems
- Integrated Master (300 ECTS)

- Diploma Thesis titled “Imperfection Detection from Photovoltaic-Panel Electroluminescence Images”.

An automated system for defect detection and categorization of photovoltaic panels was developed using deep learning. Four datasets were created and used to train Custom Convolutional Neural Networks (CNNs). Furthermore, pretrained models such as VGG-16, VGG-19, ResNet-18, and ResNet-34 were also utilized through transfer learning. The results of these models were compared with the CNNs' outcomes, evaluating their accuracy and efficiency.

Democritus University of Trace | Electrical & Computer Engineer **10/2018 – 12/2024**

- Electronics and Information Systems
- Integrated Master (300 ECTS)
- Diploma Thesis titled “Imperfection Detection from Photovoltaic-Panel Electroluminescence Images”.

An automated system for defect detection and categorization of photovoltaic panels was developed using deep learning. Four datasets were created and used to train Custom Convolutional Neural Networks (CNNs). Furthermore, pretrained models such as VGG-16, VGG-19, ResNet-18, and ResNet-34 were also utilized through transfer learning. The results of these models were compared with the CNNs' outcomes, evaluating their accuracy and efficiency.

Skills & abilities

- Microsoft Excel, Word, Latex & Power Point
- Python, Pytorch, Tensorflow
- Git, Github, Linux
- Machine Learning, Deep Learning
- Scikit Learn
- FastAPI, RESTful APIs
- Docker, Kubernetes
- Critical Thinking
- Teamwork
- Continuous knowledge enhancement

Certifications

- Python for Data Science and Machine Learning Bootcamp - *Udemy*
- The Complete Neural Networks Bootcamp: Theory, Applications – *Udemy*
- The Complete SQL Bootcamp: Go from zero to Hero - *Udemy*

Languages

- • Greece – Native
- • English – Full Professional Proficiency (C2)